Layup Laboratory

A Causal Inference and Network Science Approach to March Madness

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Abstract

This project focuses on using Network Science and Causal Inference concepts included in the MSDS-452-DL curriculum for the purpose of creating a set of diversified, competetive March Madness brackets. The first goal of this effort is creating a model that measures the causal effect of in-game events in terms of their influence on the outcome on the game. Such a model can be used as the basis for Monte Carlo simulation to determine how often one team is likely to defeat another. In recognition that some online competitions allow the user to submit dozens of brackets, the second goal is developing a strategy for building the March Madness bracket itself.

Introduction

The NCAA Division I basketball tournament, commonly referred to as March Madness, is a 63-game tournament that decides the national champion each year. As one of the most widely recognized sporting events in the United States, its arrival signals the annual tradition in which millions of fans submit their tournament predictions to casual competition pools, both in person and online. While technically consisting of 67 games, I will be excluding the play-in games from this discussion, since a majority of the popular March Madness competitions begin with the Round of 64.

The high-stakes single elimination format creates a chaotic environment in which upsets, games in which a weaker team eliminates a stronger team, are commonplace. Failure to recognize these upsets in the early rounds of the tournament tend to invalidate entire branches of bracket predictions. In recognition of the astronomically low odds that any bracket prediction is perfectly correct, generous rewards are promised to those who take on the challenge: Warren Buffet, former CEO of Berkshire Hathaway, famously offered $1 billion in 2014 (Mitchell 2024) and the SpaceX aerospace company announced that the 2025 grand prize winner would receive a trip to Mars (X Business 2025).

While the author has no hopes of claiming such rewards, he has always been fascinated with the technical challenges that are present with event forecasting. Bracketology, the term coined specifically to describe forecasting for the NCAA basketball championship, requires the analyst to make predictions at the individual game level as well as the bracket level.

Literature Review

With millions of users entering free March Madness competitions every year, the barrier for entry is low. It is a topic that both the casual sports fan and academic are willing to discuss. One may easily find herself stumbling upon interesting bracket prediction techniques from miscellaneous blogs. In one example, a blogger with the screen name *BioPhysEngr* wrote about his experience of building a bracket based on the concept of eigenvector centrality, only to be beaten by his mother picking randomly (BioPhysEngr 2012).

Traversal of binary tree data structures, on the other hand, is a richly researched topic and I anticipate finding useful academic sources for the purpose of bracket optimization.

While I continue to review the literature on both game level prediction and bracket optimization, I will briefly summarize several ideas that have caught my eye thus far.

Entropy-Based Strategies for Multi-Bracket Pools (Brill et al. 2024)

Brill et. al propose an entropy-based strategy when given the opportunity to submit multiple brackets to a competition. When allowed few brackets, optimal greedy strategies should be used, but contestants should choose more late-round upsets in order to *increase the amount of entropy in their brackets* as more are permitted, essentially diversifying their submissions.

Statistics Slam Dunk (Sutton 2024)

Uses causal inference frameworks to refute the idea of the "Hot Hand." Does not appear to delve deeply into either causal inference or network science.

SMOGS: Social Network Metrics of Game Success (Bu et al. 2019) and Social Network Analysis of College and Professional Basketball (Xu 2018)

A network can be formed by modeling the players as nodes and the number of passes between each player as directed, weighted edges. Both Xu et al. and Bu et al. suggest that teams

who pass the ball across the entire team have an advantage over teams that do not. The exception is in the Men's National Basketball Association (NBA), which rely on star players to serve as a distribution node, making a majority of the passes. The data sources currently being used (later discussed in Table 1) do not have data to this resolution, making this resource difficult to use at the current moment. The dataset used by Bu et al. does not appear to available for all NCAA Division I basketball teams, as SportsVu optical sensors are not installed in all facilities, despite being standard in theNBA. I am hopeful that player-to-player passing data is available at the college level.

Building Your Own Bracket

To use this software for bracket creation, one must set the tournament roster found in the "teams" sheet in the blank_bracket.xlsx file. If a new tournament roster is set, the user must run the round robin simulation code by executing the db_builder.ipynb notebook. Connecting to the memgraph database is not necessary. After the db_builder.ipynb code executes, all combinations of 64 teams (2,016 games in total) will have been simulated and stored in a .pkl file that can be used in the bracket_builder.ipynb.

Methods

This project is divided into two distinct lines of effort: game level prediction with causal inference and bracket optimization with various network science techniques.

Game Level Prediction

A causal model is used to understand how each event captured in game-level box scores affect the final scores of each team. Figure 1 below shows a preliminary model, which is based on the idea that team possessions lead to scoring opportunities. Here, I model the assumption that

increasing events like rebounds and steals leads to more changes in possession. Table 2 shows the preliminary average treatment effect (ATE) for each of the variables shown in the diagram. Many games can be simulated with Monte Carlo methods by randomizing the players participating on each team and the number of possessions that each team gets in each game.

Because our simulation is based on possession-level statistics, key variables in our causal model are those that influence team possessions and scoring opportunities. These include rebounds (ORB and DRB), blocks (BLK), steals (STL) and turnovers (TOV). Additionally, because none of these actually contribute directly to a team's score, their effects must be mediated through direct scoring attempts (2PA, 3PA, FTA).

Using python's DoWhy library (Sharma and Kiciman 2020), we acknowledge the validity of Figure 1 due to its resilience to refutation in the standard toolkit. While it is likely that this is not the absolute truth, our DAG is unlikely to contain placebos, additional random common causes between each variable and the respective team's score, and atypical data subsets. This method was also used for the DAG shown in Figure 3.

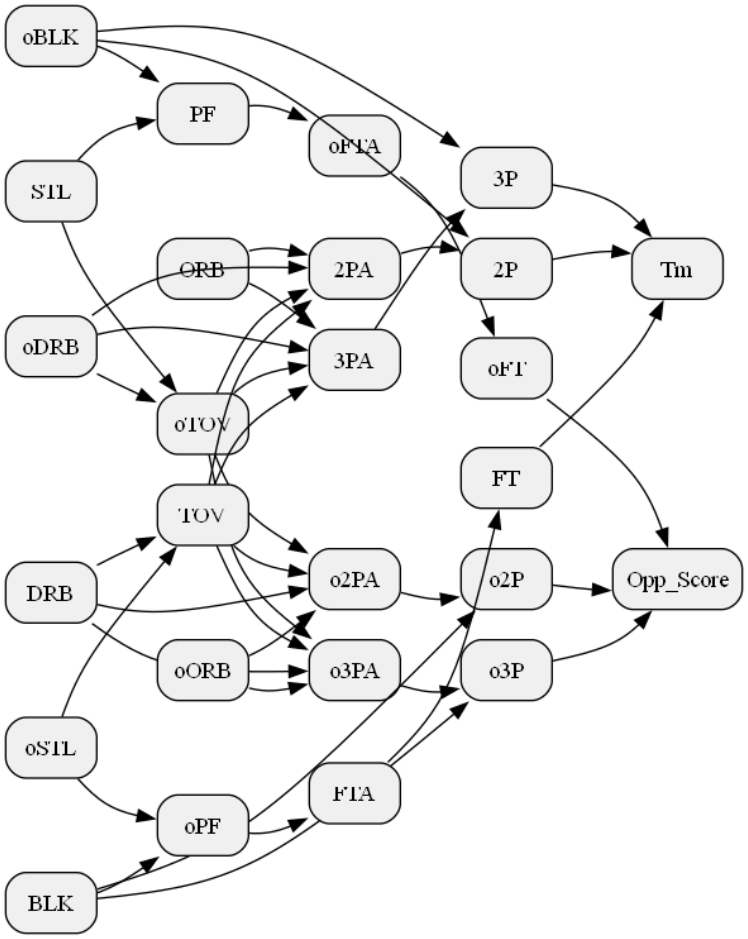

Figure 1: A preliminary DAG that can be used to estimate the point value for events that happen during a game of basketball.

Games are built around a randomly generated number of possessions and a weighted sample of players at the start of each match. Each team is given approximately the same number of possessions per game, though it varies according to each team's pace rating. Using per-possession player-level statistics, we estimate the number of events all players on the team are responsible for the decided number of possessions.

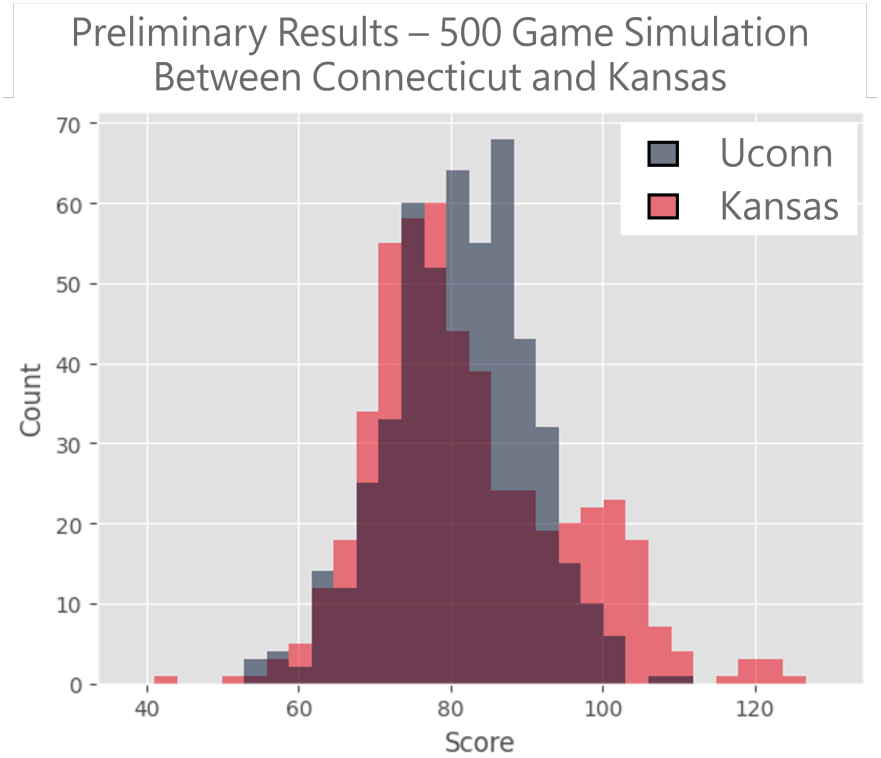

Figure 2: An overlapping histogram showing preliminary simulation score distributions between the University of Connecticut and the University of Kansas (500 games).

A benefit of the causal inference framework used in this project is the ease of explainability for each simulation. For the game above in Figure 2 above is the ability to feed the simulated game logs into a large language model and ask for a short analysis. For the 500 simulations above, gpt-5-nano (OpenAI 2024) returns the following analysis:

"Connecticut edges Kansas in the simulation, riding superior efficiency, balanced scoring, plus stronger rebounding and turnover discipline; Kansas pushes tempo but falters in efficiency and defensive consistency overall."

While this annotation process has no influence on the outcome of a round of simulations, it at least provides a fun attempt at explaining why one team might have defeated another, which might be useful in upset-situations. While Figure 2 shows the outcome of 500 games, fewer simulated games would likely be desired in high-entropy brackets, where chance plays a higher role. Thus, one way to increase the randomness in a bracket is to reduce the number of simulalted games played between each team.

Strength of Schedule

In order to broadly address changes in scoring behavior due to differences in teams' skill levels, we employ a version of the Elo rating system, borrowed from chess (Nolan 2026). With just a single rating number in which teams are awarded relatively few points for an easy win and more points awarded for a difficult decisive win, others have seen approximately a 66% success rate in previous March Madness competitions (Hanley 2024).

The graph below (Figure 3) shows the causal assumptions made during a game of basketball played with a team of a different skill level. One can measure the changes in game-level statistics as a function of elo differences. In preliminary analysis, we see that teams challenging an opponent with a rating of 100 points higher are slightly more likely to encounter additional turnovers, blocks, steals, etc., resulting in approximately 2.5 points fewer than playing teams of equal rating.

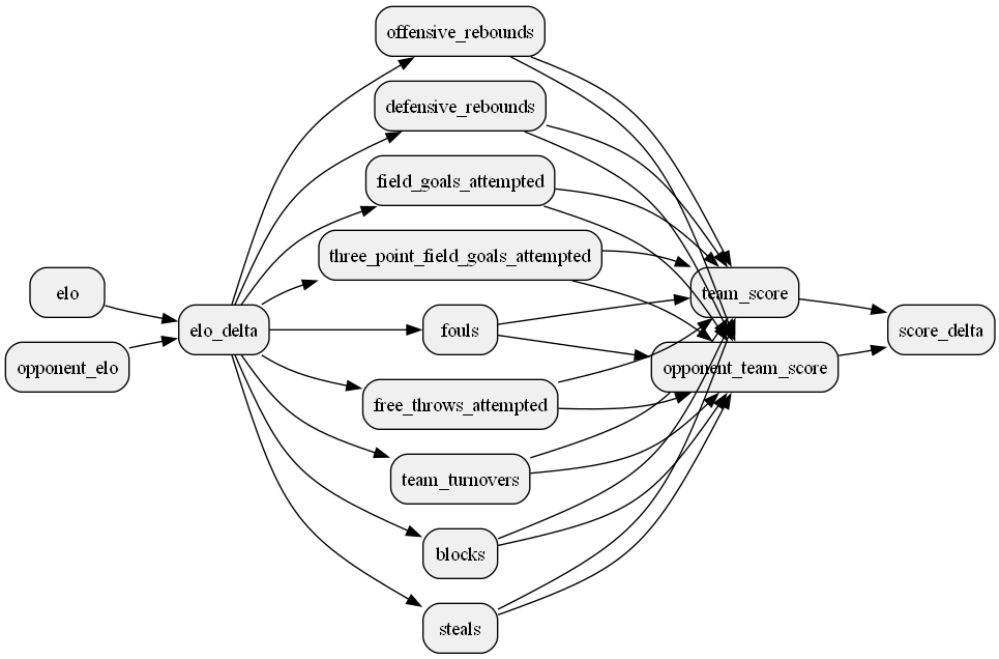

Figure 3: A directed acyclic graph showing the relationship between elo ratings and game outcomes.

Even as this project begins to draw to its close, we might recognize that additional causal models can be made to identify how different offensive and defensive variables influence a team's elo rating. A hypothesis can be formed to investigate whether offense-heavy teams perform at a higher level than their elo rating would suggest when playing against defense-poor teams, or vice versa. Offensive and defensive level performance may be latent variables that can be measured through the use of directed acyclic graphs. This data is likely available in the datasets used in this project and are a logical next step for our prediction model.

Bracket Optimization

The second line of effort in this project is optimizing the bracket performance itself. A greedy algorithm, always choosing the team with the higher chance of winning, is likely impractical when generating many brackets. Such a strategy might never account for certain

upsets. One solution would be to create one greedy version of the bracket, and create a new graph wherever our game-level simulations are uncertain of the winner. This method will hopefully prevent us from creating too many brackets where 16 seeds beat 1 seeds, but will maximize diversity in close games.

Brill et al. provide a great starting point for this problem, though I am interested in learning about different approaches as well.

Results

Two graph databases have been created for this project: one to visualize the relationships between players and teams that happened during the basketball season, and one to query pre-calculated simulations for all teams. There are 727 unique teams in the 2026 HoopR dataset, including many schoolsd that, while not division I themselves, play division I teams during the regular season. 12,476 athletes are included in this dataset. It is worth noting that athletes can transfer schools and play for multiple teams over the course of the season.

In Figure 4 below, we visualize each basketball team, the athletes that play on each team, and the outcome of all games played thus far. All edges are directed, and edges between two teams are directed and weighted according to the final difference in score.

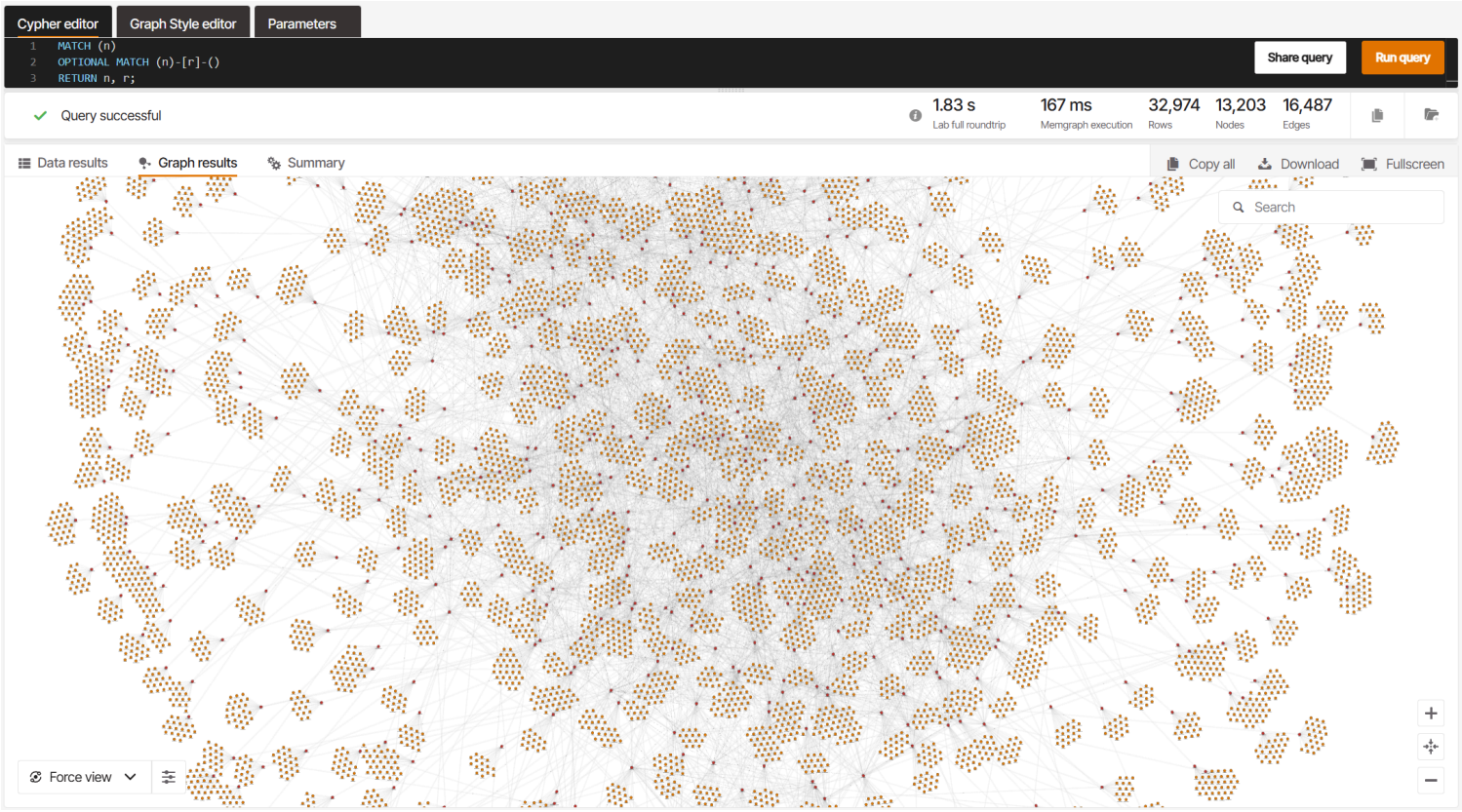

Figure 4: A sample of the 13,203 nodes and 16,487 relationships formed by teams and players that have participated in the NCAA basketball season.

In Figure 5 below, we see the results of a simulated round-robin tournament of 64 teams, to be replaced with the actual teams taking place in the NCAA Basketball Tournament once announced. This graph is created in order to drastically reduce the computations needed to produce brackets; all simulations can be performed one time each ahead of time, and only queries of the outcome will need to be performed in any given bracket permutation. While more work needs to be done on the simulations, simulations in their current state seem to favor teams with a higher ELO rating (Nolan 2026), which at least indicate forward progress.

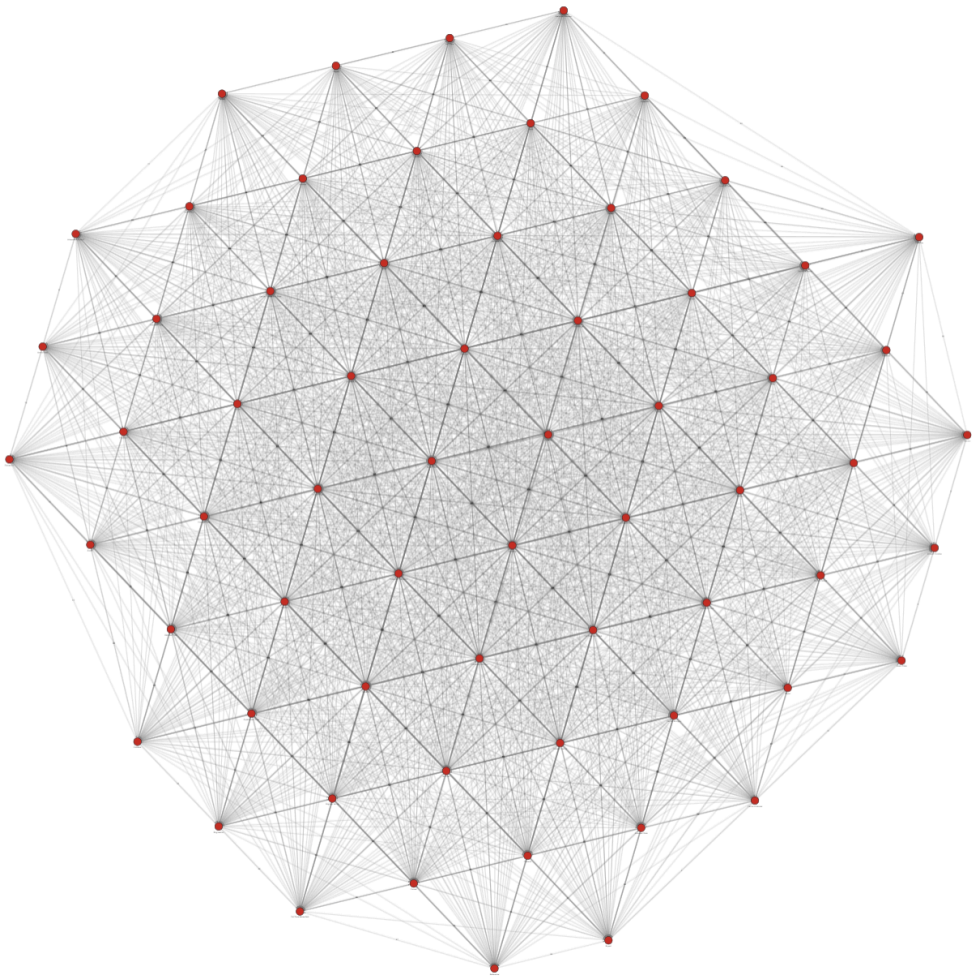

Figure 5: Simulated round-robin tournament with 64 NCAA basketball teams.

Data Sources

See Table 1 for a description of data sources used thus far. This table will be updated as the project progresses.

Source	Description
sports-reference.com (Sports Reference LLC 2026)	Team and Individual Player level data
warrennolan.com (Nolan 2026)	Team ELO ratings
hoopr.sportsdataverse.org (Gilani 2021)	Play-by-play data, schedules, and box scores

Table 1: Data Sources

Data from the sources in the table above will be used to populate causal models and a memgraph database. It may be reasonable to pre-compute a large number of games between all

possible combinations of teams and store those in a graph database prior to the bracket generation stage. Computing these outcomes beforehand will allow the bracket-building program to simply query results, rather than performing simulation calculations at runtime.

Measure of effectiveness

What should have been a simple model validation process became complicated by the nature of the data sources used in this project.

Sports Reference Data

Sports Reference, the primary source of game and player statistics in this project, maintains clear and strict webscraping guidelines for those intending to programmatically collect their data (Sports Reference LLC 2024). The use of bots and web scrapers is not strictly prohibited, but limited to 10 requests per minute. In order to responsibly use this data source, the code used in this project will make only 5 requests at a time (Sports Reference LLC, n.d.). A single request is sufficient to obtain a season's worth of games at both the player and game level for a single team. For the 365 schools that have participated in the 2025-26 Division I basketball season on Sports Reference, data could simply be scraped manually by executing code periodically. A more sophisticated schedule-based web scraping program could be created to slowly collect data from previous years, however, additional request limits may be in place. Even when attempting to stay within guidelines by making a single request per minute, the author's session was "jailed" on more than one occasion. Taking the hint, and attempting to stay respectful of the site's guidelines, only the necessary 2025-26 season data was collected.

Elo Ratings

While gathering elo rating data from warrennolan.com presented no technical hurdles, a particular interpretation of elo ratings must be adopted for use of this project. Game-by-game elo scores are not readily available from this source; only one rating is available per team, per season. Elo ratings change after each game, and finding the historical record of what the elo ratings were at the beginning of each game is a non-trivial endeavor. To complicate matters further, elo ratings continue to change during the NCAA Division I Basketball tournament. Every time a team wins or loses in the tournament, their elo rating increases or decreases accordingly. The winner of each tournament will have necessarily won 6 consecutive games, while every other team in the tournament will have lost one game. This results in the winner of each tournament gaining the most elo points throughout the tournament—in the last 5 tournaments, this elo increase has propelled the winner to also become the highest rated team in the league, regardless of whether this was the case prior to the tournament. For the purpose of this project, one must interpret a team's elo rating as a static estimate of a latent skill level.

Solution and Future Work

The solution to the issues above can be solved by simply comparing the output of simulated games to games that have already occurred By comparing the outright winner of our simulated games to the actual historical game results from the 2025-26 season, this method achieves an approximate 77.5% prediction success rate. However, the primary objective of this project is to produce March Madness brackets. The ultimate test of effectiveness will be comparing the brackets created with this program with those in the public ESPN Men's NCAA Tournament Challenge (ESPN, n.d.). Future iterations of this project could leverage network science

algorithms—such as eigenvector centrality or community detection—on the player-team graph to extract latent performance metrics and directly influence game-level predictions.

Conclusions

Amendment of objectives

Previous versions of this document included hopeful efforts to maximize bracket strategies. However, given the constraints of the current academic term and the project's extensive scope, the primary focus has been narrowed to the predictive modeling component. In order to preserve the original objectives of this projcet, a bracket-generating spreadsheet can be developed with the currently existing code base to allow users to tune brackets based on the pre-calculated round-robin graph database. This way, multiple brackets can be made, and close games (configurable) can be flipped at random. This simple technique will allow users to generate a new random bracket, without the fear of seeing all 16 seeds make it to the Final Four.

This has been done by encoding each bracket a 63 bit binary string and appending it to file. A 63 bit binary string can easily be encoded as an integer on the range of 0 to $2^{63} - 1$ for easier readability. By checking if this number already exists, we can ensure that no duplicate bracket files are made.

Event	**Team Score Effect**	**Opponent Score Effect**
2-Point Field Goal	1.998643	0.0
2-Point Attempt	1.217853	0.0
3-Point Field Goal	3.000105	0.0
3-Point Attempt	0.947277	0.0
Free Throw	1.100958	0.0
Free Throw Attempt	0.850505	0.0
Steal	1.046184	−0.779294

Event	Team Score Effect	Opponent Score Effect
Block	0.069190	−0.560639
Turnover	−0.883766	0.814006
Personal Foul	0.0	1.112083
Offensive Rebound	0.182142	0.0
Defensive Rebound	0.851554	−0.848281
Allow 2-Point Field Goal	0.0	2.002056
Allow 2-Point Attempt	0.0	0.982315
Allow 3-Point Field Goal	0.0	2.976299
Allow 3-Point Attempt	0.0	0.888978
Allow Free Throw	0.0	0.985768
Allow Free Throw Attempt	0.0	0.752256
Allow Steal	−0.901049	0.875331
Allow Block	−0.708807	0.069735
Allow Turnover	0.976682	−0.870936
Allow Personal Foul	1.004520	0.0
Allow Offensive Rebound	0.0	0.143493
Allow Defensive Rebound	−1.090965	0.688674

Table 2: Preliminary Average Treatment Effects for Player-Level Events

Results from this project can be visualized in a spreadsheet. See Figure 6 for an example output, using randomly selected teams.

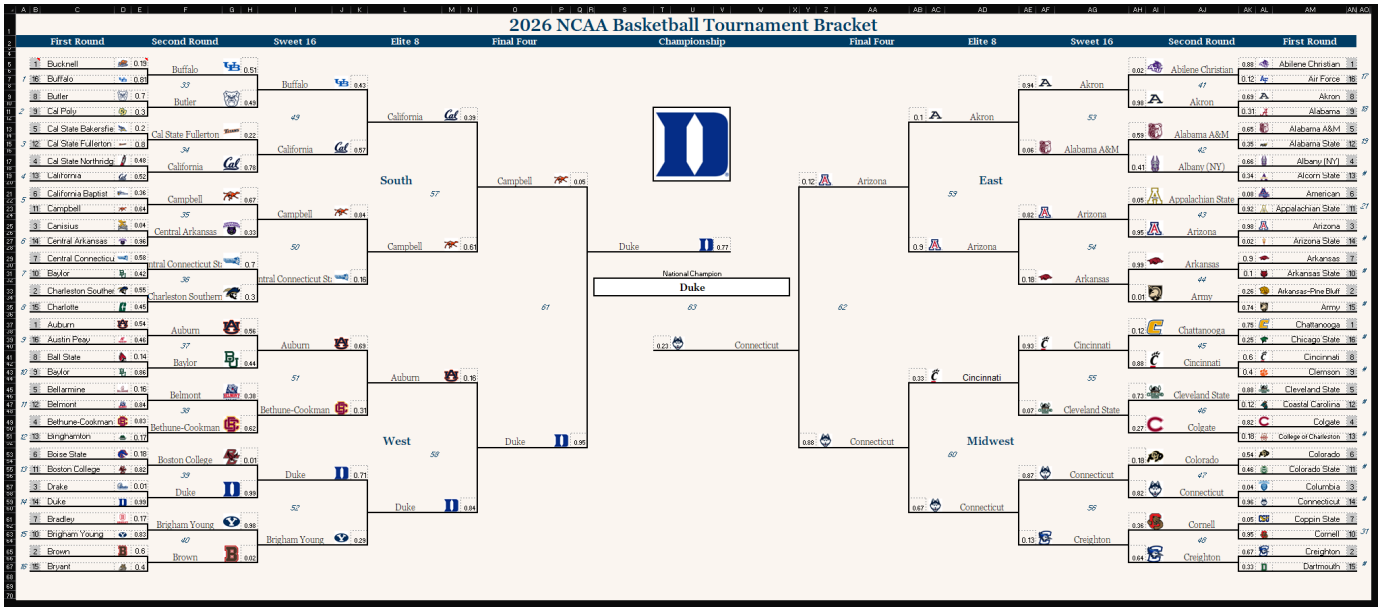

Figure 6: An example output visualization spreadsheet.

I am hopeful that the objectives defined above are attainable with the data available. While I will be entering the brackets created by this project into the official March Madness competition (details yet to be released), my only requirement for success is that this approach is better than random. ESPN tournament results will be posted in the accompanying repository as the tournament unfolds.

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